

Beyond wages, is there heterogeneity in hours worked?

Exam ID: 312

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1 Introduction

While we have seen that there are substantial disparities in wages in support of labour markets being not perfectly competitive, in this exam paper I seek to find out if there exists disparities within hours worked.

The question of whether workers are heterogeneous in hours worked is at heart a question about labour supply. It is important to understand these choices in such cases like, for example, tax reforms because "an efficient tax-benefit system should take the substantial heterogeneity in behavioral responses into account" Mastrogiacomo et al. (2017).

Furthermore, recent years have seen a degree of change in the labour markets, where more flexible hours are becoming more common. However, there has not been much known on how these options are valued by workers Mas and Pallais (2016). If indeed there are hours disparities, there could arise to adjust old policy to new realities, while having in mind where those heterogeneities come from.

To find out whether there is heterogeneity in hours worked I exploit Structure of Earning Survey of 2018 of Lithuania. I run an OLS regression with hours worked as a dependent variable, while worker and firm characteristics are independent variables. I show the how much variation my regression explains. I find that for a monthly hours worked variable, 85% of variance remains unexplained. I also show the within group standard deviations of hours worked. I group workers of the same occupation, age group, sex and education and calculate the standard deviation of their hours worked. I provide a summary table showing substantial deviation in hours worked within groups. Together, these results point to the existing heterogeneity in hours worked.

The structure of the exam paper is as follows. I begin by describing the data I am working with and provide descriptive statistics. I then state my empirical approach to the question, after which I present my results and provide explanations and my intuitions. Then I conclude my exam paper and provide a list of references and an appendix.

2 Data

My work is based on Structure of Earnings Survey, which is conducted every 4 years in Lithuania. In my case, the data is from 2018 Survey. The data features both data on worker characteristics and firm characteristic data. There are 43164 observations in total. Of this, 20688 are women and 22476 are men. 1394 People in the data set have only a basic education, 21880 people have reached secondary education, 11069 have tertiary education of up to 4 years and the rest have more than 4 years in tertiary education.

In terms of age groups, the number of workers from each age group is rather similarly distributed, the most numerous group is 50 to 59 year olds, there are 10056 of them. Of course, 60+ group is smaller due to this age group retiring.

In the data set I have two ways of determining hours worked. The first is number of hours paid in October, which combines the actual hours worked contractually with overtime hours. Another way is to use a variable "Share of a full-timer's normal hours" which is in percent and to multiply it with full time hours, which is 40. This way, the hours worked obtained is weekly and ranges from 0.216 to 40 hours worked. The distribution itself can be found in Figure 1. While my constructed Hours Weekly is less than ideal as it truncated at 40 hours a week, I will nonetheless show the results with this measure for robustness.

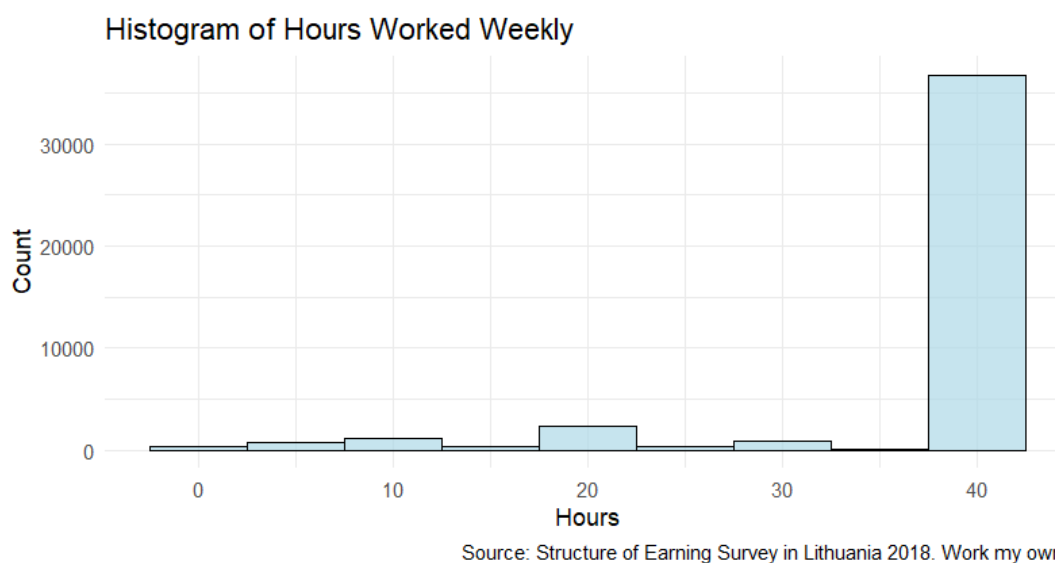


Figure 1: Histogram of Hours Worked Obtained via Share of normal hours

Alternatively, using already existing Hours Worked in October measure the hours do not cut at 40 hours weekly, but also includes those, that worked overtime. In Figure 2, you can see the Histogram of this variable. Of course there is a spike at 183-184 hours worked, which was full time worked 23 days. In Table 1 you can see this more explicitly. Also, in this table, I provide descriptive statistics on another variable, the Average Hourly Wage of October as provided in the data set.

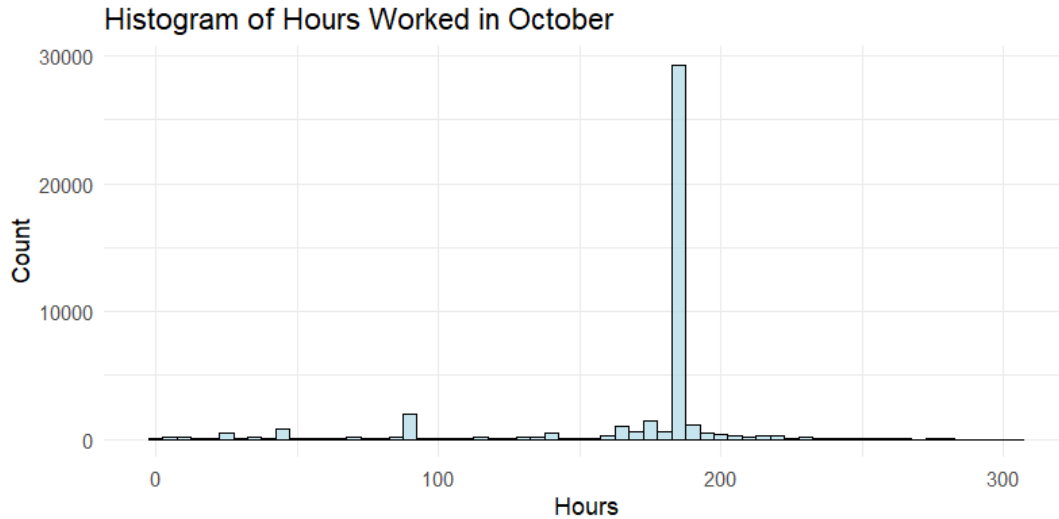


Figure 2: Histogram of Hours Worked in October

Table 1: Descriptive Statistics of Hours Worked and Wages

Measure	Hours Worked Weekly	Hours Worked October	Average Hourly Wage
Mean	36.59	168.62	5.48
Variation	78.56	1779.93	14.54
Max	40.00	306.00	116.85
Min	0.22	1.00	2.10
Q25	40.00	183.00	3.11
Q75	40.00	184.00	6.56
Mode	40.00	183.00	2.19

In terms of data related to firms, 12750 people in the data set worked in companies with fewer than 50 employees, 12890 in companies with 50-249 employees and 17524 companies with more than 249 employees. The vast majority of workers did not work under any kind of Collective pay agreement, in fact only 10431 do. In the data set, a total of 118 occupations are represented.

3 Empirical Approach

3.1 OLS

The basic approach to the research question is to run a model with hours worked as a dependent variable with worker and firm/job characteristics as independent variables: $y_i = \beta * X + \epsilon_i$.

Where y_i is the hours worked, β is vector of estimators, X is the vector of variables of worker and firm characteristics.

Such variables that I include in my regressions are age, sex, education, occupation, industry, collective payment agreement coverage, and length of service in the company. Sex, age, education, occupation, industry and collective payment agreement coverage are categorical variables.

The goal is to find the percentage of unexplained variance in hours worked. In the appendix I will also show the results for repeating this exercise with dependent variable being the average hourly wage.

3.2 Within Group Deviation

While an OLS regression could yield a result in favour there being substantial heterogeneity in hours worked, it is still necessary to provide additional evidence for it (or against it).

To this end, I will try to show deviation of hours worked within similar worker groups. That is to say, I calculate standard deviation of each group of occupation, age, sex and education. I also will only include groups with at least 50 people in it (this means 206 groups in our case), to avoid small sample sizes when calculating standard deviations. I will then report my findings from the summary statistics.

4 Results and Discussion

I run both models of hours worked with the same regressors and different dependent variables for different ways of calculating hours worked. I then extract residual variance from my regressions. I find the explained variance from the total variance of the hours worked. This, together with a percentage of variance left unexplained is reported in Table 2 for monthly worked hours and Table 3 for my constructed weekly hours.

Component	Value
Total Variance	1779.93
Explained by Observables	262.68
Unexplained (Heterogeneity)	1517.25
% Unexplained	85.24

Table 2: Decomposition of Variance of Monthly Hours Worked

Component	Value
Total Variance	78.56
Explained by Observables	11.04
Unexplained (Heterogeneity)	67.53
% Unexplained	85.95

Table 3: Decomposition of Variance of Weekly Hours Worked

As apparent from both tables, the worker and occupation leave a large part variation in hours worked unexplained, suggesting that there are other factors driving the differences in hours worked. In my view, this supports the hypothesis that there is substantial heterogeneity in labour supply choices of workers, similarly as we have seen with wages in the Applied Microeconomics course.

For my within-group standard deviation I report my results in Table 4 and Table 5. Both tables evidently show that even accounting for worker characteristics and their occupation there is substantial deviation within hours worked both if we consider monthly paid hours data and my own constructed worked hours weekly measure. With median of 33 hours and mean 34 hours per month of deviation within the same occupation, in the same age, sex and education groups, we have a strong proof that there is heterogeneity within hours worked. The same is for my measure, where the median is 6.86 hours per week of deviation and 7 hours of mean deviation. Considering that a working week usually takes 40 hours, or 8 hours per day, this result would suggest there is almost a days of difference even in the same positions for similar qualification of people.

Statistic	Value
Min.	0.32
1st Qu.	21.32
Median	33.04
Mean	34.65
3rd Qu.	47.64
Max.	83.09

Table 4: Monthly Hours

Statistic	Value
Min.	0.00
1st Qu.	4.32
Median	6.86
Mean	7.10
3rd Qu.	10.35
Max.	15.83

Table 5: Weekly Hours

Table 6: Within-Group Standard Deviation of Hours Worked

These points of evidence both point to there existing substantial hours disparities (as seen with wages). As in that case, I would argue, this may arise from worker-firm preferences. Although, I do not test this in any way in this paper. Here, too, mobility costs could play a role. For example, instead of moving, you could take up less hours. Or monopsony power, where few employers have power not only to set wages, but hours too.

Although, this is a very primitive measure, but a simple correlation calculation between hours worked and average hourly wage seems to be non-existent. The correlation between monthly worked hours and average hourly wage is 0.019 and between weekly hours and average hourly wage 0.018. This would not suggest that the disparities in hours are driven by differences in wages.

5 Conclusions, Implication and Limitations

I would conclude that the results that I have obtained point to there existing meaningful disparities in hours worked. Which would mean that a story that is told by classical labour market framework is not quite correct. Given heterogeneity in hours worked, a higher wage might not induce someone to work longer hours due to their preferences, or a person might prefer to work long hours and wage might matter less. All of this would mean that we should be careful on how we design and apply policy when thinking about welfare of workers.

However, this exam paper is limited due to only having cross sectional data for one year and one country. The data on hours worked does not include unpaid work or work done in the shadow sector.

Furthermore, OLS is a simple method, so anything related to it should always be taken with a grain of salt. Also, this paper does not fully exhaust every possible way of testing for heterogeneity, which also limits the strength to which I can proclaim my results.

References

- Mas, A., Pallais, A., 2016. Valuing alternative work arrangements. NBER Working Paper URL: <http://www.nber.org/papers/w22708>, doi:10.3386/w22708.
- Mastrogiacomo, M., Bosch, N., Gielen, M., Jongen, E., 2017. Heterogeneity in labour supply responses: Evidence from a major tax reform. Oxford Bulletin of Economics and Statistics 79. doi:10.1111/obes.12168.

6 Appendix

6.1 Wage case

Below in the table I provide the decomposition of Variance of Wages, here, consistent with both results from hours regressions and to things we have seen in the course, there are large disparities in wages.

Component	Value
Total Variance	14.54
Explained by Observables	4.04
Unexplained (Heterogeneity)	10.50
% Unexplained	72.23

Table 7: Decomposition of Variance of Wages